

ILE D'ORLEANS, QC, Canada

WMO: 714240

Lat: 46.9967N

Lon: 70.8083W

Elev: 12

StdP: 14.69

Time Zone: -5.00 (NAE)

Period: 90-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.8	-5.6	-21.9	1.6	-10.2	-16.8	2.2	-3.9	36.8	22.0	35.1	20.0	14.2	230	0.740

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.4	82.2	68.2	79.1	66.5	76.1	65.2	71.2	77.1	69.5	75.0	67.8	73.5	15.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.3	107.6	74.4	67.4	100.9	73.0	65.7	94.7	71.0	35.1	77.5	33.6	75.1	32.2	73.5	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 36.5	31.9	28.0	DB	-15.1	87.2	4.1	2.3	-18.0	88.8	-20.5	90.2	-22.8	91.4	-25.8	93.1	(4)
(5)			WB	-15.6	74.1	3.6	3.1	-18.2	76.4	-20.3	78.2	-22.4	80.0	-25.0	82.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.4	14.5	16.5	26.9	38.7	52.4	62.9	66.8	66.5	58.1	46.1	34.1	23.2		
	DBStd	20.03	11.20	10.83	8.57	6.18	6.84	6.63	4.94	4.90	6.33	6.65	7.76	10.00		
	HDD50	4624	1101	937	715	345	49	1	0	0	8	158	480	830		
	HDD65	8513	1566	1357	1180	790	396	117	37	40	222	586	927	1295		
	CDD50	1835	0	0	0	4	123	387	520	511	251	37	2	0		
	CDD65	250	0	0	0	0	5	53	92	86	14	0	0	0		
	CDH74	1107	0	0	0	2	47	242	400	354	62	0	0	0		
	CDH80	195	0	0	0	0	9	46	75	55	10	0	0	0		
(14) Wind	WSAvg	12.8	13.8	12.9	13.2	13.6	13.7	12.3	11.5	11.7	12.0	13.0	12.9	13.4		
(15) Precipitation	PrecAvg	45.0	3.3	2.8	2.8	3.3	3.4	4.4	5.1	4.6	4.0	4.2	3.5	3.7		
	PrecMax	53.3	5.5	4.7	4.7	7.3	5.0	8.6	9.0	7.3	5.9	8.9	5.1	6.1		
	PrecMin	35.1	1.1	1.9	0.9	0.4	1.7	1.9	2.6	1.0	2.2	1.1	1.5	1.7		
	PrecStd	5.6	1.3	0.8	0.9	1.8	1.0	1.4	1.6	1.5	0.9	1.9	0.9	1.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.4	42.1	47.0	63.9	79.3	85.1	86.3	84.8	80.7	69.1	60.7	43.8		
		MCWB	37.7	37.9	38.8	53.1	63.9	68.8	70.2	68.3	67.5	57.8	56.5	39.6		
	2%	DB	35.4	37.4	41.9	54.8	72.4	80.5	81.6	81.2	74.5	62.9	51.9	39.2		
		MCWB	33.5	34.3	37.0	46.2	58.3	65.9	68.3	68.4	65.0	56.0	47.5	36.2		
	5%	DB	32.3	34.3	39.0	49.5	67.0	76.2	78.4	78.0	70.4	58.2	47.5	36.9		
		MCWB	30.8	32.1	35.1	42.2	56.3	64.0	66.5	66.9	63.0	52.4	43.2	34.3		
	10%	DB	29.4	31.8	36.6	46.1	62.5	72.8	75.2	75.1	67.4	54.8	43.8	35.0		
		MCWB	27.8	30.1	33.2	40.4	54.2	62.6	65.5	65.8	60.6	50.0	40.1	33.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.5	39.7	42.1	55.6	66.0	72.1	74.6	72.4	70.2	62.8	57.0	41.3		
		MCDB	39.4	41.4	45.2	59.2	78.5	79.8	82.2	79.2	75.5	65.8	62.3	43.0		
	2%	WB	33.7	34.8	38.1	47.1	61.0	68.7	71.9	71.0	67.7	57.1	48.7	36.9		
		MCDB	34.9	36.8	40.5	52.9	68.6	75.2	77.9	76.8	72.6	62.3	50.7	38.6		
	5%	WB	30.8	32.6	35.9	43.6	58.0	67.2	69.2	69.6	64.7	53.7	43.6	34.9		
		MCDB	31.9	34.2	38.7	48.2	65.3	73.4	74.6	74.9	69.5	57.3	46.6	36.5		
	10%	WB	27.6	30.0	33.8	41.2	54.9	65.3	67.0	67.8	62.0	50.9	40.6	33.2		
		MCDB	28.8	31.7	36.1	45.2	61.4	71.6	72.3	73.2	66.2	54.0	43.4	34.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	13.0	13.1	11.0	10.5	13.5	13.6	13.4	13.7	12.7	10.3	9.2	10.4		
		MCDBR	16.0	12.8	12.6	17.0	22.7	19.4	18.2	17.5	17.6	15.5	13.8	10.2		
	5% WB	MCWBR	15.8	11.5	9.3	11.0	12.2	9.6	8.9	9.0	10.7	11.1	12.7	9.3		
		MCWBR	15.5	12.4	12.3	15.2	19.9	16.7	14.9	14.9	15.8	14.5	13.6	9.6		
(40) Clear-Sky Solar Irradiance	taub	taub	0.262	0.285	0.326	0.357	0.367	0.402	0.417	0.395	0.373	0.341	0.310	0.288		
		taud	2.338	2.261	2.227	2.331	2.382	2.303	2.285	2.347	2.402	2.479	2.452	2.313		
	Ebn at Noon	Ebn at Noon	265	280	283	285	285	274	268	269	265	258	245	237		
		Edh at Noon	23	30	37	37	37	40	40	36	31	25	21	21		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	442	730	1108	1409	1605	1741	1699	1477	1157	677	390	298		
	RadStd	RadStd	60	89	89	119	167	112	171	94	96	56	46	29		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page